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Brac University

Semester: Summer 2024

Course Code: CSE460

VLSI Design

Section: 01-08

Set
02

Assessment: *Midterm Exam*

Duration: 1 hour 30 minutes

Date: 13 July, 2024

Full Marks: 45

Instructions: You MUST answer **Question 1**. You are also required to answer any two of **Questions 2–4**.

■ Question 1 of 4 [CO2] [15 marks]

You are to design a **finite state machine** (FSM) that automatically turns off an AC after 4 hours of continuous use. Let's assume you provide a clock to your FSM that gives a positive edge after every hour. The AC provides an input w to your machine in each cycle. Your machine has to provide an output $z = 1$ in the 4th hour if the AC is on for the previous 3 consecutive hours. Otherwise, the machine provides $z = 0$

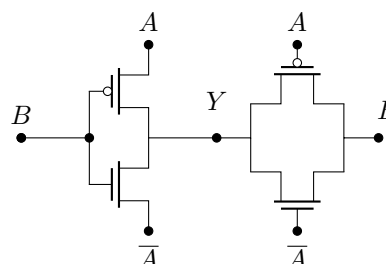
- [5 marks] **Identify** the type of FSM for the above input-output relationship and **design** the state diagram for the machine.
- [5 marks] **Design** the binary-encoded state assigned table of the FSM.
- [5 marks] **Determine** the logic expressions of the next state and output variable z from the state assigned table.

■ Question 2 of 4 [CO1] [15 marks]

There are four sensors, A, B, C, and D, that control the air conditioning (AC) in a room. Each sensor enters a **HIGH** state when it detects a temperature above a preset threshold. The AC will turn **ON** when the majority of the sensors are in the **HIGH** state. If there is an equal number of sensors in the **HIGH** and **LOW** states, the state of the AC (either ON or OFF) is **indeterminate**, i.e., in don't care condition. Answer questions (a-c) below based on this description.

- [2 marks] **Write** down the truth table for the system. Describe all possible states of the sensors (A, B, C, D) and their corresponding AC output states.
- [5 marks] **Draw** the Karnaugh map (K-map) for the system based on the truth table in part (a). Derive the logical expression that governs the system from the K-map.
- [5 marks] **Draw** the CMOS logic circuit for the logical expression derived in part (b). Specify the number of MOSFETs required to implement your design.

- [3 marks] **Fill** in the **Y** column of the table on the right in your answer script, assigning either 0 or 1 based on the logic output of the adjacent circuit. In this context, 0 represents a low voltage of 0V, and 1 represents a high voltage of 5V.



A	B	Y
0	0	
0	1	
1	0	
1	1	

■ Question 3 of 4 [CO1] [15 marks]

The parameters for the MOSFETs in the circuits depicted in **Figures 3a** and **3b** are detailed in the **adjacent table**. Additionally, both the pMOS and nMOS devices in these circuits have the same oxide layer thickness, permittivity, and aspect ratio. These are specified as $t_{ox} = 10 \text{ \AA}$, $\epsilon_{ox} = 3.9 \times 8.854 \times 10^{-14} \text{ F/cm}$ and $W/L = 50$, respectively.

Parameters for Fig. 3a

nMOS	$\mu_n = 1000 \text{ V/cm}^2$	$V_{tn} = 1 \text{ V}$
pMOS	$\mu_p = 500 \text{ V/cm}^2$	$V_{tp} = -1 \text{ V}$

Parameters for Fig. 3b

Both MOSFETs	$\mu_n = 2\mu_p$	$V_{tn} = V_{tp} = 0 \text{ V}$
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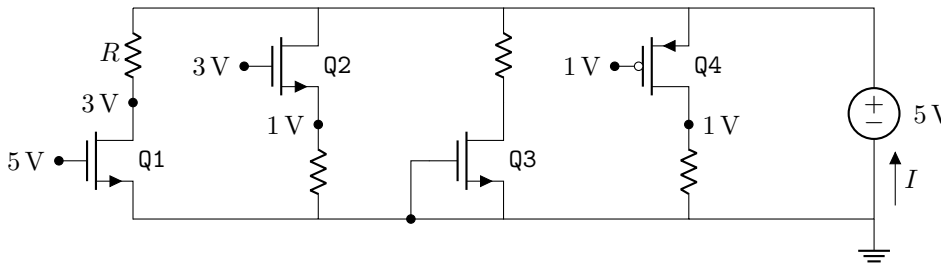


Figure 3a

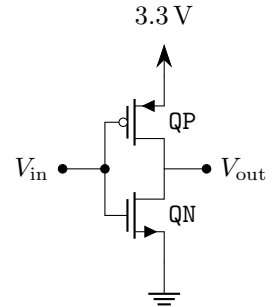


Figure 3b

Answer questions (a-d) based on **Fig. 3a**.

- [2 marks] Find β_n and β_p of the MOSFETs.
- [3 marks] Determine the operating regions of each of the transistors in **Fig. 3a**.
- [3 marks] Find the current I , through the 5 V voltage source.
- [2 marks] Find the value of R .

Answer the next question based on **Fig. 3b**.

- [5 marks] Find the switchover/inversion voltage V_{inv} of the inverter circuit in **Fig. 3b**.

■ Question 4 of 4 [CO1] [15 marks]

The MOSFET in the circuit of Fig. 2 has the following parameters:

$\mu_n = 450 \text{ V/cm}^2$, $V_t = 0.2 \text{ V}$, $t_{ox} = 10 \text{ \AA}$, $\epsilon_{ox} = 3.9 \times 8.854 \times 10^{-14} \text{ F/cm}$ and $W/L = 20$.

- [2 marks] Find β and gate capacitance per unit area of the MOSFET.
- [3 marks] If the MOSFET is in saturation mode with $I_d = 1 \text{ mA}$, calculate V_g .
- [3 marks] What should be an appropriate range of the values for R_1 for the MOSFET to be in saturation mode?
- [2 marks] Find out the operating mode of the MOSFET with proper verification if $R_1 = 3.8 \text{ k}\Omega$. use the value of V_g obtained in part (b).

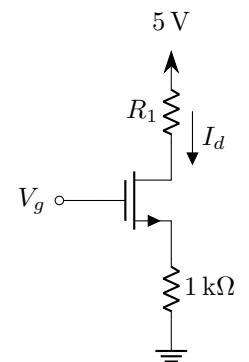


Figure 2

- [5 marks] In the **adjacent figure**, the supply voltage (V_{DD}) is 5 V. The threshold voltages for the pMOS and nMOS transistors are both set at $V_{tn} = |V_{tp}| = 0.1 \text{ V}$. Determine the voltages V_1 , V_2 , V_3 , and V_4 as well as the output voltages V_X and V_Y . [Z is also a valid output.]

